

Linear Search

```
LAB1.cpp
LAB1.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int n;
7      cout << "Enter the size of array: ";
8      cin >> n;
9
10     int arr[n];
11     cout << "Enter " << n << " elements:" << endl;
12     for (int i = 0; i < n; i++)
13     {
14         cin >> arr[i];
15     }
16
17     int key;
18     cout << "Enter element to search: ";
19     cin >> key;
20
21     int comparisons = 0;
22     int position = -1;
23
24     for (int i = 0; i < n; i++)
25     {
26         comparisons++;
27         if (arr[i] == key)
28         {
29             position = i;
30             break;
31         }
32     }
33
34     if (position != -1)
35     {
36         cout << "\nElement found at index: "
37              << position << endl;
38         cout << "Number of comparisons: "
39              << comparisons << endl;
40
41         if (position == 0)
42         {
43             cout << "Case: Best Case" << endl;
44             cout << "Time Complexity: O(1)" << endl;
45         }
46         else if (position < n - 1)
47         {
48             cout << "Case: Average Case" << endl;
49             cout << "Time Complexity: O(n)" << endl;
50         }
51         else
52         {
53             cout << "Case: Worst Case" << endl;
54             cout << "Time Complexity: O(n)" << endl;
55         }
56     }
57     else
58     {
59         cout << "\nElement not found." << endl;
60         cout << "Number of comparisons: "
61              << comparisons << endl;
62         cout << "Case: Worst Case" << endl;
63         cout << "Time Complexity: O(n)" << endl;
64     }
65 }
```

Enter 3 elements:

1 2 3

Enter element to search: 3

Element found at index: 2

Number of comparisons: 3

Case: Worst Case

Time Complexity: O(n)

PS C:\Users\shobh\Downloads\Coding\APL LAB> [

Binary Search

```
Binary_Search.cpp > main0
1  #include <bits/stdc++.h>
2  using namespace std;
3
4  int binarySearch(int arr[], int n, int key, int &comparisons)
5  {
6      int low = 0;
7      int high = n - 1;
8
9      comparisons = 0;
10
11     while (low <= high)
12     {
13         int mid = low + (high - low) / 2;
14
15         comparisons++;
16
17         if (arr[mid] == key)
18             return mid;
19
20         if (arr[mid] < key)
21             low = mid + 1;
22         else
23             high = mid - 1;
24     }
25
26     return -1;
27 }
28
29 int main()
30 {
31     int n, key;
32     int comparisons;
33
34     cout << "Enter number of elements: ";
35     cin >> n;
36
37     int arr[n];
38
39     cout << "Enter " << n << " sorted elements:\n";
40     for (int i = 0; i < n; i++)
41     {
42         cin >> arr[i];
43     }
44     cout << "Enter element to search: ";
45     cin >> key;
46
47     int result = binarySearch(arr, n, key, comparisons);
48
49     if (result != -1)
50     {
51         cout << "\nElement found at position: " << result + 1;
52         cout << "\nNumber of comparisons: " << comparisons;
53     }
54     else
55     {
56         cout << "\nElement not found.";
57         cout << "\nNumber of comparisons: " << comparisons;
58     }
59
60     cout << "\n\nComplexity Analysis:";
61     cout << "\nBest Case : O(1)";
62     cout << "\nAverage Case : O(log n)";
63     cout << "\nWorst Case : O(log n)";
64
65     return 0;
66 }
```

```
> cd "c:\Users\shobh\Downloads\Coding\APL Lab\" ; if ($?) { g++ Binary_Search.cpp -o Binary_Search } ; if ($?) { .\Binary_Search
}
Enter number of elements: 5
Enter 5 sorted elements:
3 5 7 9 10
Enter element to search: 3

Element found at position: 1
Number of comparisons: 2

Complexity Analysis:
Best Case : O(1)
Average Case : O(log n)
Worst Case : O(log n)
```